1 In the following question, the digits C_3, C_2, C_1 refer to the three least significant digits of your CID (College Identifier). For example, if your CID is shown on your swipe card as '00123456' then $C_3 = 4$, $C_2 = 5$, $C_1 = 6$.

a A CPU manufacturer's data-sheet quotes performance in terms of how many instructions are completed in one second.

i) What is the standard three- or four-letter acronym used to describe this metric, and what does it stand for?

MIPS = Millions of Instructions per second

or

IPS = Instructions per second

Reading: 1 minute; thinking: 0 minute; answer: 1 minute. Total: 2 minutes

Marks: **2**

ii) Give one advantage and one dis-advantage of this performance metric.

Advantage: very easy to measure. Advantage: changes of MIPS in one architecture correlate with observed performance. Disadvantage: can't easily compare between different ISAs. Disadvantage: different compilers may achieve lower MIPS but higher observed performance.

Reading: 1 minute; thinking: 1 minute; answer: 1 minute. Total: 3 minutes

Marks: **2**

b A program is compiled and run in a system with a five-stage pipelined MIPS-1 CPU running at 80MHz. The CPU pipeline is identical to the one seen in lectures, and includes an instruction and data cache. Empirical tests show the system completes $(1 + C_1) \times 10^7$ instructions for each second the program runs.

i) State your value of C_1, and calculate the IPC for this program.

Getting them to state C_1 allows us to check for follow-through errors if they get the wrong digit.

The IPC is $(1 + C_1)/8$, for whatever value of C_1 they have.

Reading: 4 minute; thinking: 1 minute; answer: 1 minute. Total: 6 minutes
Reading includes the time take to work out what C_1 is.

Marks: **2**

ii) Choose one of the following categories to describe the achieved IPC, and justify your choice:

A: The CPU is achieving optimal performance.

B: The program experiences a typical number of stalls.

C: The program causes a high number of cache misses.

D: The achieved IPC is impossible in a five-stage MIPS.

Classifications are:

- *A :* $C_i = 7$, $IPC = 1$, *so the CPU is achieving optimal performance, with no stalls. This is not typical.*
- *B :* $C_i = 6, 5, 4, 3$, $IPC = 0.88, 0.75, 0.62, 0.5$, *a typical level of stalls, possibly due to control/data hazards, or cache misses.*
- *C :* $C_i = 4, 3, 2, 1, 0$, $IPC = 0.62, 0.5, 0.38, 0.25, 0.12$, *this can only be due to a lot of cache misses, as control/data hazards can't cause this much slow-down by themselves.*
- *D :* $C_i = 8, 9$, $IPC = 1.12, 1.25$, *this is impossible to achieve in a pipelined MIPS.*

Different classifications may be accepted if they are supported by a reasonable justification.

Reading: 3 minute; thinking: 2 minute; answer: 2 minute. Total: 7 minutes

Marks: **2**

c Assume a memory system with an L1 and L2 cache connected to a DDR RAM. The hit time of the L1 cache is $h_{L1} = 1 + C_2$ cycles, and the miss-rate of the L1 cache is $m_{L1} = (1 + C_3)/20$. The DDR RAM has a fixed hit time of 34 cycles per access, and never misses.

i) State the values of the constants h_{L1} and m_{L1}, then give an expression for the Average Memory Access Time (AMAT) of the system that substitutes numerical values for all known constants.

Our overall AMAT is $h_{L1} + m_{L1}(h_{L2} + m_{L2}h_D)$.

Getting them to state h_{L1} *and* m_{L1} *allows us to check for follow-through errors if they get the wrong digit out of their CID.*

This is then substituted through for the specific parameters chosen, e.g. for example:

- $C_2 = 0$, $C_3 = 0$, *AMAT=*$h_{L2}/20 + 17/10m_{L2} + 1$

- $C_2 = 2$, $C_3 = 8$, *AMAT*=$h_{L2}/20 + 153 * m_{L2}/10 + 3$,

Every combination of C_2 and C_3 has been calculated using Python+sympy, and checked to make sure the results are sensible (e.g. not negative or something).

The answer can be given as fractions, or using decimals.

Reading: 4 minute; thinking: 2 minute; answer: 2 minute. Total: 8 minutes
Reading time includes time taken to work out C_2 and C_3.

Marks: **3**

ii) Use your knowledge of the typical relationships between L1 and L2 cache performance to provide a numeric lower-bound on the AMAT of this system. State the knowledge you rely on, and any other assumptions.

We know that in general:

- *The miss rate of an L2 cache is lower than an L1 cache.*
- *The hit time of an L2 cache is higher than an L1 cache.*

So a lower-bound can be got by assuming that $m_L2 = m_L1$ and $h_L2 = h_L1$. Substituting through results in a variety of values from 1.19 cycles through to 34.2.

For example:

- $C_2 = 0$, $C_3 = 0$, *AMAT*=$h_{L2}/20 + 17/10 m_{L2} + 1$,
  min_AMAT=$143/120 = 1.1917$
- $C_2 = 9$, $C_3 = 9$, *AMAT*=$h_{L2}/2 + 17 * m_{L2} + 10$,
  min_amatp=$175/6 = 29.16667$

Every combination of C_2 and C_3 has been checked.

Answers could uses other reasonable rules of thumb, such as $m_{L2} = 1 - (1 - m_{L1})^2$, or $h_{L2} = h_{L1} + 1$, as long as they are stated and the general principles that $h_{L2} \geq H_{L1}$ and $m_{L2} \geq m_{L1}$ is taken.

Reading: 1 minute; thinking: 3 minute; answer: 1 minute. Total: 5 minutes

Marks: **3**

d The EEE department wishes to purchase a large number of new workstations for student use, and has received quotes from twelve potential suppliers. Each supplier's quote proposes a different system, and contains a variety of common quantitative information about the system's cost, performance, and capabilities.

i) Propose a quantitative metric which reduces each quote to a single number m, where lower values of m are better. Your metric should rely on at least two clearly identified pieces of quantitative information from the quotes. Briefly justify why this metric makes sense from the point of view of purchasing workstations.

Any metric should take into account price, and combine it with some measure of computational power. We can't easily run custom applications on all sixteen machines, so need to pick standard benchmarks that can be run by vendors. A reasonable benchmark for the department would be SpecFP, as the department uses a lot of floating-point intensive applications. To take into account the number of cores, we could either use modern spec which scales to many cores, or run parallel version. The metric is then simple total spec score divided by price. (Many other metrics are possible, e.g. involving power, possibly memory, and so on).

The inputs to the metric should be clearly identified (e.g. cost per unit, result of a particular benchmark), and should clearly be a formula that can be applied.

An example metric could be:

$$m = c/p \tag{1}$$

where c is the unit cost, and p is the performance achieved in SpecFP.

Any metric should go in the obvious direction with each parameter (e.g. it should go up as cost goes up, and down as performance goes up). There may be constants included which need to be tuned.

Reading: 2 minute; thinking: 3 minute; answer: 2 minute. Total: 7 minutes

Marks: **2**

ii) Invent parameters for two different hypothetical systems, and use your metric to show which one should be bought.

Machine 1: $c = 800$, $p = 3100$, $m = 0.258$

Machine 2: $c = 700$, $p = 3500$, $m = 0.2$

Machine 2 has the better metric.

Reading: 1 minute; thinking: 1 minute; answer: 1 minute. Total: 3 minutes

Marks: **2**

iii) Give an example of a system which would do well according to your metric, but in fact would not be appropriate for the EEE department.

Any machine which violates "obvious" assumptions would work. For example, a system which consumed 10kW of power if the metric does not consider power, or a system which only has 64MB of RAM if the metric does not consider RAM.

Alternatively, some seriously skewed system could work: e.g. the unit cost is 1M pounds, even though the performance is extremely high.

Reading: 1 minute; thinking: 2 minute; answer: 1 minute. Total: 4 minutes
Question: 44 minutes

Marks: **2**

The four parts carry, respectively, 20%, 20%, 30%, and 30% of the marks.

2 A designer has decided to create a "compressed" 16-bit instruction format for MIPS. This format will be used in addition to the standard 32-bit instruction format, and binaries will be allowed to contain a mixture of 16-bit and 32-bit instructions. 32-bit instructions will be fetched and processed as normal. 16-bit instructions will be expanded into a full 32-bit instruction in the ID stage, with the EX and MEM stages will remain unchanged.

a What is the main advantage of supporting 16-bit instructions?

More instructions can be fit into the same amount of space. This means more instructions can fit in the icache; less bandwidth needed to refill the icache; smaller program memory needed.

Reading: 5 minute; thinking: 1 minute; answer: 1 minute. Total: 7 minutes

Marks: **3**

b Give an example of a 32-bit MIPS instruction which cannot be encoded as a single 16-bit instruction, regardless of the 16-bit encoding chosen.

`j 0x56678` : *this contains more than 16-bits of information, and so cannot be put into one instruction.*

`addi r18, r24, 0xF234` : *The combination of registers and immediates could not be encoded in any general-purpose 16-bit encoding.*

Reading: 1 minute; thinking: 2 minute; answer: 1 minute. Total: 4 minutes

Marks: **2**

c Describe the main changes needed to the instruction-fetch stage to support a mix of 16-bit and 32-bit instructions.

The program counter needs to be extended by 1-bit to support half-word addresses. There needs to be a new flag or control signal which describes whether the program-counter advances by 2 bytes or 4 bytes.

The memory path can stay the same, as it will just fetch the same 32-bit value twice from cache.

Reading: 1 minute; thinking: 3 minute; answer: 2 minute. Total: 6 minutes

Marks: **3**

d Assume a program binary contains a sequence of three arithmetic instructions. The first and third instruction are 32-bit, while the second instruction is 16-bit,

and the instructions are placed back-to-back in memory. Assuming the first and second instructions execute correctly, what problem will occur on the third instruction?

If we assume the first instruction executes correctly, then it must be aligned to a word boundary. The following 16-bit instruction is then also word aligned, but finishes on a half-word. That means that the final 32-bit instruction is not word-aligned, which is illegal in MIPS, so there would be an exception or fault.

Reading: 2 minute; thinking: 2 minute; answer: 1 minute. Total: 5 minutes

Marks: **2**

e It has been decided that the 16-bit J instruction will follow the form of the 32-bit J instruction, but with a reduced-size immediate field. What is the set of addresses a 16-bit J instruction can jump to? Explain any assumptions needed.

We need to decide whether the new instruction can jump to word-aligned locations, or whether it can now jump to half-word aligned instructions. If we assume that it is still word-aligned, then the immediate size is now 10 bits, and the bottom two bits will be zero. So the set of locations is `concat((pc+4)[31:12],imm[9:0],zero[1:0])`*, where pc is the current program counter and imm is the 10 bits from the instruction. Ignoring the '+4' is also fine.*

It could also be defined for half-word locations, with half the total jump distance.

The students may discuss whether jumps imply exiting 32-bit mode or not, but this is not required.

Reading: 2 minute; thinking: 3 minute; answer: 2 minute. Total: 7 minutes

Marks: **3**

f Propose a new encoding for R-type instructions in the 16-bit format. Assume that there is still a 6-bit op-code, and the op-code has the same meaning as the normal 32-bit instruction op-code.

We need to decide how to fit both control flags and registers into the available 10 bits. Traditional function+shift would take up all of these bits, so we will remove shift and restrict the function to just 4 bits. This leaves 6 bits. We could either use these as three 2-bit fields, or two 3-bit fields. Taking the second option we'll choose:

- `instruction[15:10]` : *opcode*
- `instruction[9:7]` : *register 1*

- `instruction[6:4]` : *register 2*
- `instruction[3:0]` : *function*

Because we only have two operands, the first register will be both a source and a destination.

There are many sensible alternatives, as long as they fit within 16-bits and have reasonable semantics that allow most operations to complete.

Reading: 1 minute; thinking: 4 minute; answer: 3 minute. Total: 8 minutes

Marks: **4**

g Describe a tradeoff you had to make when designing your new R-type encoding.

There is a tradeoff between the size of the function code and the number of possible register operands - the chosen method allows for 16 function codes, but only 8 possible registers per operand. We could have chosen fewer possible functions, but allowed a larger set of possible operands.

There is a tradeoff between number of operands versus available registers per operand. Assuming that many instructions are of the form $a = op(a,b)$ allows for greater flexibility in the choice of a and b, but means that $a = op(b,c)$ cannot be implemented in one instruction.

Reading: 1 minute; thinking: 2 minute; answer: 3 minute. Total: 6 minutes
Question: 43 minutes

Marks: **3**

The seven parts carry, respectively, 15%, 10%, 15%, 10%, 15%, 20%, and 15% of the marks.